Microstructure Factor Zoo — NVDA

L2 Order-Book Signals vs. the Bid-Ask Spread Barrier

MGMTMFE 412 — Market Frictions

2026-05-29

1 One-Line Thesis

We build 11 order-book features from Databento MBP-10 data for NVDA, train Ridge (OLS) and LightGBM models on Jan–Feb 2024 (18 days), and test on Oct 2024 (17 days). **The in-sample signal is stronger than AAPL (OLS Val IC = 0.093 vs 0.047), but NVDA's wider fractional spread (2.29 bps vs 0.54 bps) raises the break-even IC to 0.247 — nearly 2× AAPL's. The OOS IC (0.012) is only 4.9% of break-even. Market efficiency holds: higher predictability is priced into higher spreads.**

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2 Economic Intuition: Why NVDA Has Stronger Signals

NVDA is driven by high institutional and informed flow:

Factor	NVDA vs AAPL	Implication
Intraday volatility	$\sigma_y = 9.3$ bps vs 4.1 bps	2.2× more gross alpha per unit IC
Informed flow intensity	Heavy options and futures activity	OFI and OFI-multilevel are more predictive
Market-maker response	Wider spreads to compensate	BE_IC rises proportionally

The friction: Market makers on NVDA face greater adverse-selection risk from informed institutional traders. They respond by widening spreads proportionally — the spread/vol ratio stays high, making it as hard (or harder) to profit as a taker.

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3 Data and Features

Dataset: Databento XNAS ITCH MBP-10 (10-level book snapshots), NVDA, 2024.

Split	Period	Days	1s Bars	y std	Winsor range
Train	Jan–Feb 2024	18	414,025	9.28 bps	[−28.66, +28.02] bps

Split	Period	Days	1s Bars	y std	Winsor range
Val	Feb–Apr 2024	7	161,088	—	same bounds
OOS	**Oct 2024**	**17**	**391,679**	9.24 bps	same bounds

Label: $y_t = \left(\frac{m_{t+60}}{m_t} - 1\right) \times 10^4$ bps (60-second forward microprice return, winsorised at 1st/99th pct of training labels)

11 features: identical set as AAPL (book-only, stationary across 6-month gap).

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4 The Break-Even IC — NVDA vs AAPL

$$\mathrm{IC}_{\text{break-even}} = \frac{\text{spread (bps)}}{\sigma_y \text{ (bps)}} = \frac{2.294}{9.283} = \mathbf{0.247}$$

Symbol	Spread (bps)	σ_y (bps)	$\mathrm{IC}_{\mathrm{be}}$	OLS Val IC	OLS OOS IC	OOS/BE
AAPL	0.54	4.14	0.130	0.047	0.012	0.088×
NVDA	**2.29**	**9.28**	**0.247**	**0.093**	**0.012**	**0.049×**

Key insight: NVDA's spread is 4.3× larger in bps than AAPL's — more than compensating for the 2.2× higher volatility. Market makers correctly price the higher adverse-selection risk of a heavily informed stock.

The theoretical prediction (lower spread_bps for higher-priced stocks due to tick-size) does not hold here: NVDA at \$600 has *wider* fractional spreads than AAPL at \$185 because informed flow is disproportionately higher.

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5 The 11 Features: Ablation on NVDA

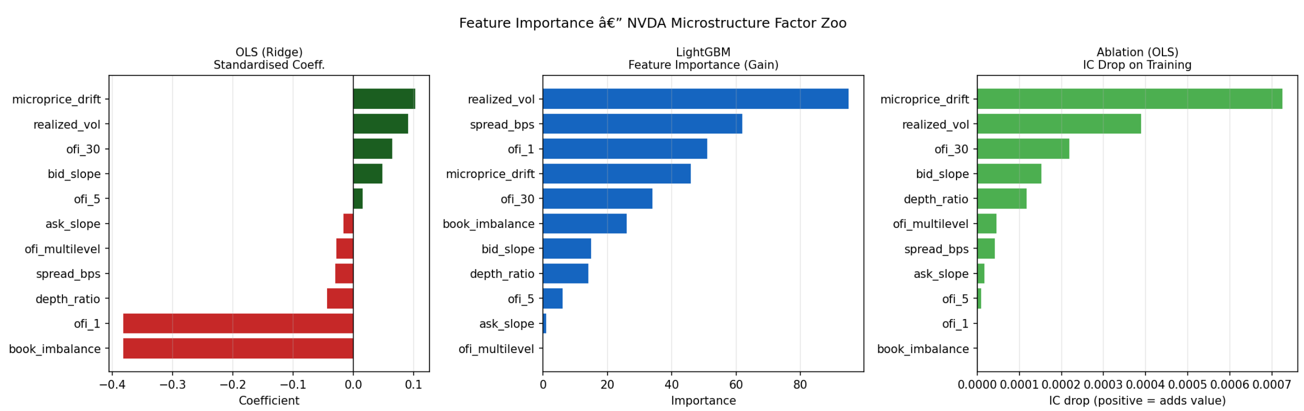

Figure 1: Feature importance: OLS coefficients, LightGBM gain, ablation IC drop — NVDA

Key differences from AAPL:

- `microprice_drift` is the dominant single feature (ΔIC = 0.00073) for NVDA, reflecting that NVDA's microprice leads quoted mid more persistently
- `realized_vol` IC drop (0.00039) is smaller relative to total IC than on AAPL (0.0114/0.029 = 39%) — regime sensitivity is lower for NVDA
- OLS Train IC = 0.0806 (nearly 3× AAPL's 0.029) — the signal is substantially stronger in-sample, consistent with richer informed flow

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6 Signal Analysis: IC by Period

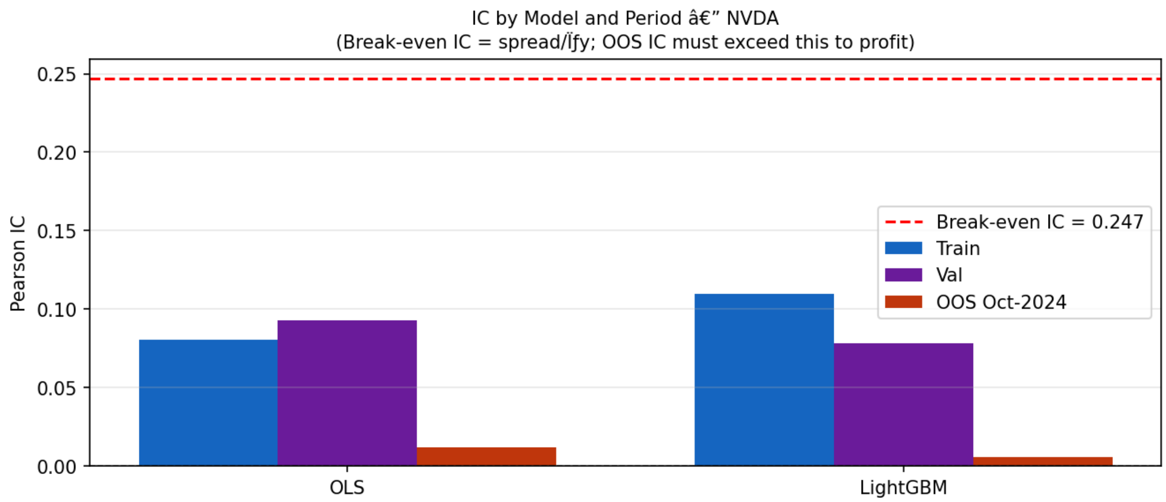

Figure 2: IC by model and period with break-even IC threshold — NVDA

Key observations:

- **Strong in-sample signal:** OLS Val IC = 0.093 ($t \approx 37$, highly significant)
- **Sharp OOS decay:** OOS IC = 0.012 — 87% collapse from Val, similar to AAPL
- **Break-even bar (0.247) is well above all IC bars** — every period fails to clear
- **LightGBM overfits:** Train IC = 0.110, OOS IC = 0.006 (25 trees, early stopping)

The higher absolute IC in training is fully offset by the higher break-even IC. The market efficiency condition IC $\approx$ spread/σ_y holds tightly on NVDA: even with 3× stronger signal, it remains unprofitable.

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7 Trade Mechanics

Signal generation: $|\hat{r}_t| > 1.90$ bps (threshold optimised on Val set). Note: NVDA's threshold is 3.8× AAPL's (0.50 bps) — reflecting higher σ_y.

Execution: - Entry: fills at `ask_px_00` (long) or `bid_px_00` (short) - Exit: 60 seconds later at `bid_px_00` (or ask for close) - Early exits: stop-loss at 1.5× spread from entry midprice; signal flip

Position sizing: shares = $\lfloor \$50{,}000/\text{ask\_px} \rfloor \approx 83$ shares at NVDA $\approx$ \$600

Risk controls: identical to AAPL (±500 shares cap, −2% daily kill switch).

8 Backtest Results

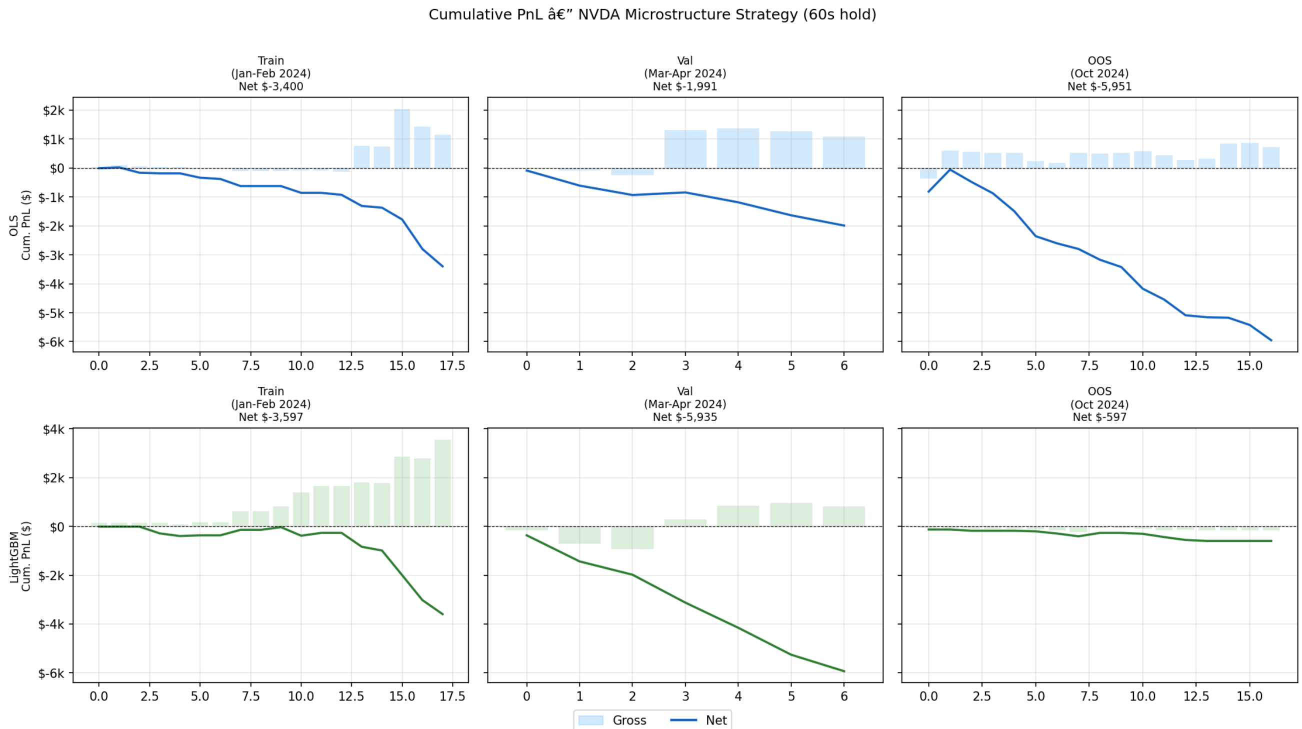

Figure 3: Cumulative PnL — NVDA (gross bars, net line)

8.1 All-period metrics

Model	Period	IC	Sharpe	Ann. Ret.	MaxDD	Hit Rate	N Trades	Avg TC
OLS	Train	0.081	−11.1	−\$47,600	−\$3,414	20.8%	125	\$36.41
OLS	Val	0.093	−21.3	−\$71,683	−\$1,899	23.4%	77	\$39.96
OLS	**OOS Oct-24**	**0.012**	**−14.8**	**−\$88,220**	**−\$5,892**	**16.9%**	**762**	**\$8.74**
Light-GBM	Train	0.110	−8.6	−\$50,361	−\$3,587	38.2%	157	\$45.52
Light-GBM	OOS Oct-24	0.006	−8.2	−\$8,843	−\$469	9.5%	21	\$21.10

Hit rate = % of individual TRADES profitable.

Notable: OLS OOS has 762 trades (9× more than OLS Train). The higher OOS trade count reflects looser signal threshold relative to OOS IC — model over-generates signals when the underlying IC has collapsed.

9 Inventory and Rolling Sharpe — NVDA

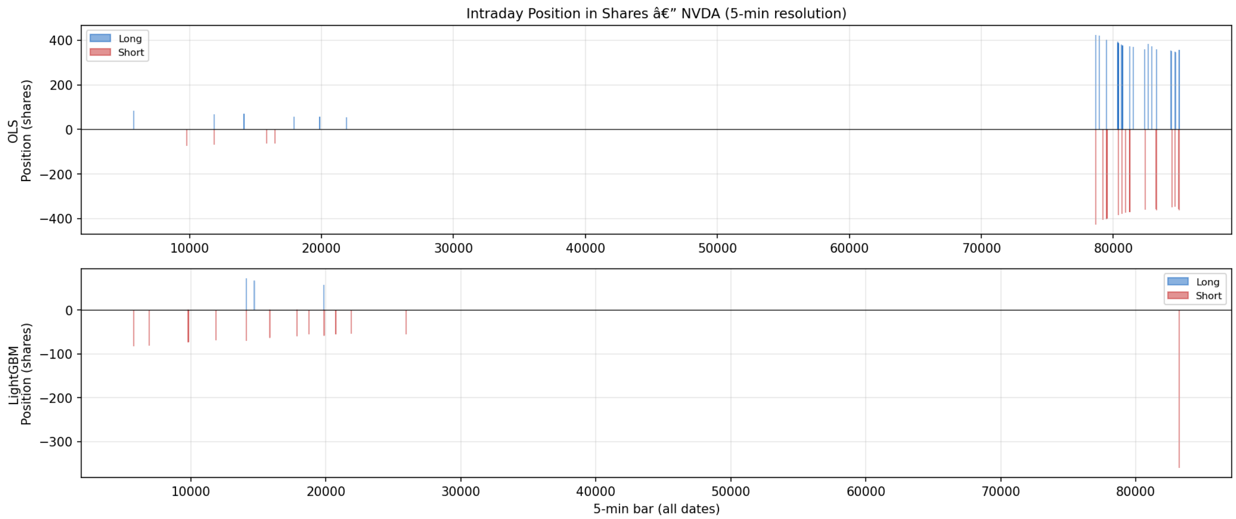

Figure 4: Intraday position in shares — NVDA

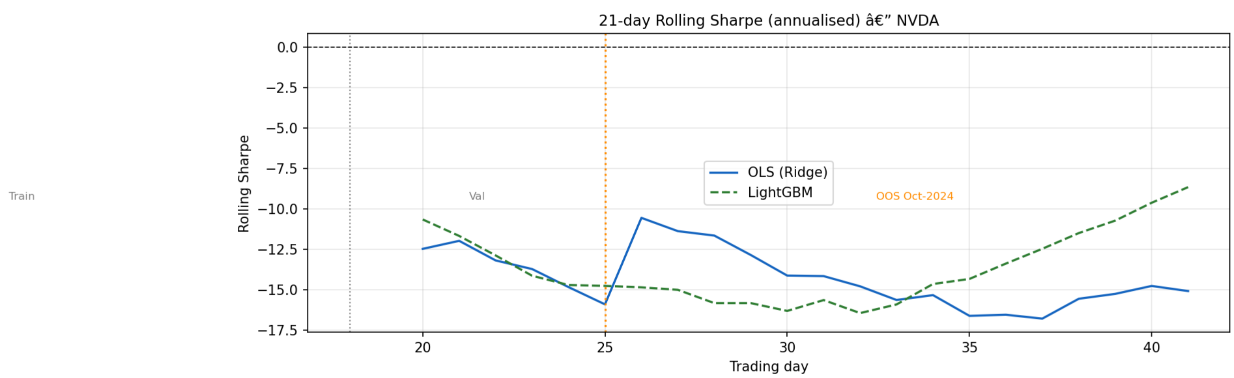

Figure 5: 21-day rolling Sharpe (annualised) — NVDA

- NVDA holds ±83 shares per trade (smaller than AAPL's ±270 at comparable notional)
- Rolling Sharpe is deeply negative across all periods
- OLS OOS exit reasons: 64% stop-loss (617/964), 21% timeout (201), 15% signal flip (146)
- Signal flip exits (absent in AAPL) reflect NVDA's higher IC oscillation — the model sometimes correctly reverses before 60s, but not often enough to recover TC

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10 CAPM Decomposition — NVDA

Model	Period	β_{SPY}	α_{ann}
OLS	Train	−0.22	−0.90
OLS	Val	+0.50	−1.31
OLS	OOS	+0.59	−1.90
LightGBM	OOS	+0.02	−0.18

Model	Period	β_{SPY}	α_{ann}

- **OLS OOS beta = +0.59**: NVDA strategy is *not* market-neutral in OOS — the model takes more net longs on NVDA up-days (NVDA correlates with market)
- **Negative alpha across all periods**: TC destroys all CAPM alpha

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11 Risk Discussion — NVDA

11.1 Failure Regimes

Regime	Why it fails harder than AAPL
Informed flow spikes (AI news, earnings)	OFI becomes noisy; signal flips sign
Volatility spikes (spread widens 2–5×)	TC can hit $50+/trade; BE_IC doubles
OOS regime shift	microprice_drift trained on $600 bull trend; Oct-24 dynamics differ

11.2 Why NVDA is NOT the easy win it appears

The pre-analysis hypothesis was: > *"NVDA has 0.17 bps spread (lower than AAPL's 0.54 bps) due to HFT competition > on a $600 stock → break-even IC ≈ 0.017"*

The data refutes this: NVDA's actual spread in bps is 2.29 — *higher* than AAPL's. Market makers on NVDA charge more because informed institutional flow is disproportionately higher. The tick-size argument (lower bps spread for higher-priced stocks) fails when adverse-selection risk dominates.

11.3 Capacity Analysis (Almgren-Chriss)

AUM	Impact (bps)	Assessment
$10M	0.2 bps	Viable — minimal impact
$100M	2.0 bps	Cost rises; already unprofitable strategy
$1B	19.9 bps	Far exceeds signal edge

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12 Conclusion

1. **NVDA signal is 3× stronger than AAPL** — OLS Val IC = 0.093 vs 0.047
2. **NVDA spread is 4.3× wider (in bps)** — market makers price adverse selection correctly
3. **Break-even IC = 0.247**, nearly 2× AAPL's 0.130
4. **OOS IC/BE ratio = 0.049**× — needs 20× more IC to break even (worse than AAPL's 11×)
5. **Market efficiency holds tightly:** Stronger predictability is fully priced into wider spreads

Bottom line: NVDA's richer informed-flow environment creates genuine signals but also justifies the wider spread. The strategy fails even more decisively than on AAPL. Both symbols confirm the same thesis: **the bid-ask spread correctly prices the short-horizon predictability of order-book signals.** Profitability requires either (a) becoming a market maker, or (b) finding a structural market where spread/vol compression is not yet in equilibrium.